

SOVEREIGN SYSTEM SECTOR REPORT

Autonomous Mother Algorithm Performance Ledger & AI Advisory

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TARGET INSTRUMENT: R_75 Volatility Model

CORE PERFORMANCE METRICS

Total Settlements:	700	Calculated Profit Factor:	7.08
Win Accuracy:	77.3% (541W / 159L)	Max Peak Drawdown:	0.1%
Cumulative PnL:	\$421.25	Avg. PnL per Trade:	\$0.60

EQUITY CURVE PROGRESSION



PERFORMANCE MILESTONES (100-TRADE SEGMENTS)

SEGMENT	WIN RATE	NET PnL	REFINEMENT EFFICIENCY
Trades 1-140	82.1%	\$21.65	+99% [OK]
Trades 141-280	100.0%	\$161.13	+120% [OK]
Trades 281-420	93.6%	\$166.20	+112% [OK]
Trades 421-560	54.3%	\$51.38	+65% [OK]
Trades 561-700	56.4%	\$20.89	+68% [OK]

SOVEREIGN NARRATIVE ADVISORY

Cognitive Strategy Proposal & Adaptive Intelligence Logs

1. Executive Telemetry Critique

An analysis of the 700-trade settlement run reveals highly anomalous, structural imbalances. While the macro metrics suggest an institutional-grade system, micro-level breakdown exposes severe vulnerability to structural regime shifts and execution drag.

...

Metric	Value
Total Trades (N)	700
Win Rate (WR)	77.29%
Profit Factor (PF)	7.08
Max Peak Drawdown (DD_max)	0.10%
Cumulative PnL	\$421.25
Expected Value (EV) / Trade	\$0.60

...

Mathematical Decomposition

- Gross Performance:** With $PF = 7.08$ and $Net\ PnL = \$421.25$, we derive:
$$\text{Gross Losses (GL)} = \frac{\text{Net PnL}}{\text{PF} - 1} = \frac{421.25}{6.08} \approx \$69.28$$
$$\text{Gross Profits (GP)} \approx \$490.53$$
- Average Trade Metrics:**
$$\text{Average Win } (W_{\mu}) = \frac{\$490.53}{541} \approx \$0.91$$
$$\text{Average Loss } (L_{\mu}) = \frac{\$69.28}{159} \approx \$0.44$$
$$\text{Risk-to-Reward Ratio } (R:R) = \frac{W_{\mu}}{L_{\mu}} \approx 2.07:1$$

Risk Engineering Verdict

- Sliver-Thin Edge Margin (Commission Ruin Risk):** An EV of $\$0.60$ per trade on 700 iterations is a critical vulnerability. In a live production environment, standard exchange fees, clearing fees, and bid-ask slippage (estimated at $\$0.05$ to $\$0.15$ bps per side) will degrade this edge to near-zero or negative. The system is currently over-fit to zero-friction environments.
- Unrealistic Drawdown Profiles:** A maximum drawdown of 0.1% across 700 trades suggests either extreme under-leverage (capital underutilization) or a highly curated, non-representative historical sample (low-volatility regime bias). It indicates a lack of tail-risk exposure (black swan events) within the test window.
- Asymmetry Paradox:** Achieving a 77.3% win rate alongside a 2.07 positive reward-to-risk ratio is statistically anomalous in liquid markets. This indicates the logic is exploiting a highly persistent, micro-mean-reverting pricing inefficiency that is highly sensitive to latency.

2. Regime Suitability & Behavioral Patterns

The current configuration parameters optimize specifically for Low-Volatility, Range-Bound (Mean-Reverting) Regimes with low directional drift.

```
...
[Volatility Expansion] -> System enters trade at BB 2.25
|
+--> [ADX > 25 (Trend)] ----> High stop-out risk via ATR 3.15
|
+--> [ADX < 25 (Range)] ----> Rapid mean-reversion (High Win Rate)
...
```

Behavioral Weaknesses by Market State

* **Trend-Intensity Breakdown (ADX > 25):** The configuration uses Bollinger Bands (`bbStd: 2.25`) and RSI thresholds (`30/70`) as entry triggers. In an expansionary, trending regime (ADX > 25), these boundaries act as "falling knife" traps. The `atrStopMultiplier` of `3.15` is wide enough to let trending moves run deeply against the position before liquidation, while the `maxTicksInTrade: 120` time-stop acts as a forced realization of loss at peak trend extension.

* **Volatility Compression (Quiet Ranges):** This is where the model harvested its 77.3% win rate. When volatility stays compressed inside the 2.25 standard deviation envelope, the mean reversion engine functions perfectly. However, the system is structurally unprepared for volatility regime shifts (e.g., transition from low-volatility compression to high-volatility expansion).

* **Confluence Redundancy:** A `minConfluenceScore` of `3` combined with standard Bollinger Band and RSI thresholds suggests the model requires near-simultaneous boundary breaches. This restricts the trade frequency, leading to the low absolute PnL (\$421.25) across a high trade count (700).

3. Calibrated Recommendations

To scale this system from a fragile micro-scalper to an institutional-grade, robust execution algorithm, we must adjust the parameters to increase the average trade size (\$EV\$) and protect against trend-expansion risk.

Recommended Parameter Re-calibration

```
```json
{
 "rsiOversoldThreshold": 25,
 "rsiOverboughtThreshold": 75,
 "bbPeriod": 20,
 "bbStd": 2.50,
 "maxTicksInTrade": 75,
 "minConfluenceScore": 3,
 "atrStopMultiplier": 2.10,
 "regimeAdxThreshold": 20
}
```

## # Parameter Justification

`rsiOversold/Overbought` [30/70 -> 25/75]: Tightening these boundaries filters out premature counter-trend entries. This reduces trading frequency but shifts execution to high-probability exhaustions, increasing the average win size ( $W_{\mu}$ ).

`bbStd` [2.25 -> 2.50]: Widening the envelope from 2.25 to 2.50 standard deviations ensures entries occur only at true statistical extremes (98.76% of distribution under normal conditions). This insulates the book from false breakout entries.

`maxTicksInTrade` [120 -> 75]: Reducing the holding time-stop minimizes exposure to trend persistence. If mean reversion does not occur quickly within 75 ticks, the structural thesis is invalidated; exiting early prevents tail-end losses.

`atrStopMultiplier` [3.15 -> 2.10]: Tightening the stop-loss prevents catastrophic "fat-tail" losses. Combined with the wider entry envelope (2.50 SD), the model can afford a tighter stop, skewing the mathematical expectancy toward a safer distribution.

`regimeAdxThreshold` [25 -> 20]: Lowering the ADX threshold forces the system to lock out mean-reversion entries at an earlier stage of trend development, preventing the algorithm from fighting emerging momentum.

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## SOVEREIGN RISKS SENTINEL SYSTEMS

Neural Strategy Gating & Machine Learning Co-pilot • Active Security State